Jie Zhu

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Research Interests **Agentic AI, VLMs, Reinforcement Learning, VQA, Multimodal**

PUBLICATIONS

Selected Publications

- **FusionAgent: A Multimodal Agent with Dynamic Model Selection for Human Recognition.**
  Jie Zhu, Xiao Guo, Yiyang Su, Anil Jain, and Xiaoming Liu. CVPR26. [Paper] [Project] [☆8]
  Keywords: Agentic AI, MLLMs, Refinforcement Learning
- **Can Textual Reasoning Improve the Performance of MLLMs on Fine-grained Visual Classification?**
  Jie Zhu, Yiyang Su, and Xiaoming Liu. CVPR26 Findings. [Paper] [Project] [☆44]
  Keywords: MLLMs, Reinforcement Learning, Visual Understanding
- **A Quality-Guided Mixture of Score-fusion Experts Framework for Human Recognition**.
  Jie Zhu, Yiyang Su, Minchul Kim, Anil Jain, and Xiaoming Liu. ICCV25. [Paper] [☆18]
  Keywords: Multimodal, MoE
- **DepthAgent: Towards Better Universal Depth Estimation via Sample-wise Expert Selection.**
  Jie Zhu, Girish Chandar Ganesan, and Xiaoming Liu. ICLR27 submission. [Paper]
  Keywords: Agentic AI, Depth Estimation, MLLMs
- **Seeing with Tools, Reasoning with Language: Recasting Vision as Structured Scene Text for Driving VQA.**
  Jie Zhu, Xingpeng Sun, Yibo Yang, Xin Li, Wenbin He, Xiaoming Liu, and Liu Ren. ICLR27 submission.
  Keywords: Driving VQA, Agentic AI
- **LTDriveQA: Benchmarking Spatiotemporal Intelligence in Multi-View Long-Tail Driving Scenarios.**
  Jie Zhu, Haolan Xu, Zhihao Zhang, et al. ICLR27 submission.
  Keywords: Driving VQA, Benchmark

Others

- **SapiensID 2.0: Aligning Human Recognition Foundation Models with Human Perception.**
  Yiyang Su, Jie Zhu, Feng Liu, Anil Jain, and Xiaoming Liu. NeurIPS26. [Paper]
  Keywords: VLMs, Human Recognition
- **Subtoken Vision Transformer for Fine-grained Recognition.**
  Jie Zhu, Ivy Zhang, Minchul Kim, and Xiaoming Liu. ACCV26 (**Oral**). [Paper] [☆]
  Keywords: Image Tokenization, ViT
- **ATM: Action Temporality Modeling for Video Question Answering.**
  Junwen Chen, Jie Zhu, and Yu Kong. ACM MM23. [Paper][☆4]
  Keywords: VQA, Action Understanding
- **Person Recognition at Altitude and Range: Fusion of Face, Body Shape and Gait.**
  Liu Feng, ..., Jie Zhu, et al. TPAMI. [Paper]
  Keywords: Multimodal, Biometrics
- **Fairness-Sensitive Policy-Gradient Reinforcement Learning for Reducing Bias in Robotic Assistance.**
  Jie Zhu, Mengsha Hu, Amy Zhang, and Rui Liu. IEEE ROMAN24. [Paper]
  Keywords: Reinforcement Learning, Fairness
- **VLG-Drive: Unifying Vision-Language and Geometric Priors for Autonomous Driving.**
  Junge Zhang*, Jie Zhu*, Yanqi Jia, Xiangyu Xu, et al. ICLR27 submission.
  Keywords: Autonomous Driving, VLMs
- **AgentMine: Tool-Grounded Agentic Reinforcement Learning for Autonomous Driving Scenario Mining.**
  Xingpeng Sun, Jiajing Guo, Jie Zhu, Yibo Yang, et al. ICRA27 submission.
  Keywords: Autonomous Driving, VLMs

EDUCATION

Ph.D. in Computer Science, UNC at Chapel Hill, Advisor: Prof. Xiaoming Liu. Aug 2026 – Apr 2028
Ph.D. in Computer Science, Michigan State University, 4.0/4.0, Advisor: Prof. Xiaoming Liu. Aug 2023 – Apr 2026
MS in Computer Science, George Washington University, 3.9/4.0, Advisor: Prof. Abdou Youssef. Sep 2021 – May 2023
BS in Computer Science, Northeastern University (China), 3.2/4.0, Advisor: Prof. Yuan Liu. Aug 2016 – Jun 2020

RESEARCH PROJECTS

RSI-Exam Benchmark

- Contributed the `budgeted_depth_expert_routing` task to RSI-Exam, where an agent iteratively develops a router from RGB inputs, camera metadata, expert depth predictions, and visible-set feedback, then generalizes to a **sealed set** by selecting and fusing five frozen depth experts under an accuracy–cost trade-off.

Seeing with Tools, Reasoning with Language: Recasting Vision as Structured Scene Text.

- Proposed **DriveTLM**, a training-free framework that decouples visual perception from language reasoning by converting vision-tool outputs into *structured scene text*, enabling VLMs/LLMs to reason without visual tokens.
- **+21.6 points** average gain on DriveBench and up to **+10.2 Lingo-J points** on LingoQA over direct visual input, with particularly strong gains for smaller models.

DepthAgent: Adaptive Expert Selection for Universal Depth Estimation.

- Proposed DepthAgent, a GRPO-trained VLM agent that performs sample-wise depth expert selection and fusion via multi-turn tool use, achieving strong performance across diverse camera domains, especially on hard samples.
- **+3.9**, **+1.8**, and **+2.7** δ_1 points on hard perspective, fisheye, and panoramic samples over the strongest single expert, with **0.97s** on fast inference per sample.

FusionAgent: A Multimodal Agent with Dynamic Model Selection for Human Recognition.

- Proposed **FusionAgent**, an agentic MLLM that leverages *multi-turn ReAct reasoning* and GRPO for dynamic model selection, introducing an *Anchor-based Confidence Top-k (ACT)* fusion for adaptive multimodal integration.
- **+13.2%** on CCVID, **+7.5%** on MEVID, and **+17.7%** improvement on LTCC benchmark over SoTA, with the same efficiency (2.6s for CoT, 1s for fast answer per sample).

Can Textual Reasoning Improve the Performance of MLLMs on Fine-grained Visual Classification?

- Discovered the performance degradation in CoT under zero-shot evaluation and multiple training paradigms. We uncover a central paradox: the degradation induced by CoT is largely driven by the reasoning length in FGVC.
- **+3.6%** on average accuracy across FGVC benchmarks via the proposed ReFine-RFT with ensemble, accuracy-centric rewards while constraining reasoning, and *MRN* for optimization to enhance the RFT training.

WORK EXPERIENCE

Bosch USA United States
Research Intern May 2026 – Aug 2026

- Developed **DriveTLM**, a tool-augmented driving VQA framework that integrates perception and temporal tools with LLM reasoning through structured scene text.
- Built **LTDriveQA**, an *event-centered* driving VQA benchmark with **2,889 human-reviewed questions over 719 long-tail multi-view clips**, featuring *question auditing*, and a scalable *recursive question-generation* pipeline.

Inter-American Development Bank United States
AI Analytics Consultant Jun 2023 – Aug 2023

- Engineered multilingual web scraping pipelines (50+ media outlets) using BeautifulSoup and Scrapy.
- Built a **ChatGPT-powered dashboard** for summarization and trend analysis of 10,000+ daily text/video news.

HONORS & AWARDS

Graduate Tuition Fellowship (3 out of 74) Aug 2022
Third Prize Scholarship (30 out of 302) Sep 2018 – Jul 2020

SKILLS & SERVICES

Reviewer: ICLR27; NeurIPS26; TCSVT; TPAMI; FG (2024-2026); IJCBLRR24
Skills: Python, PyTorch, Hugging Face, LaTeX, GitHub, MuJoCo, Unity (AR/VR), HTML
Grants: Contributed to grant writing for UniDisplayAgent (Samsung), Object ReID (IARPA), and End-Gen (IARPA).
Others: Maintained the CVLab website for 3 years.

SELECTED PATENTS

CN113194348B, "Virtual human lecture video generation method, system, device and storage medium". Jul. 2022.
CN113192161B, "Virtual human image video generation method, system, device and storage medium". Oct. 2022.